Yiming **Huang**

IMPERIAL · DEPARTMENT OF COMPUTING

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Research

My research interests primarily focus on developing novel topological deep learning methods for understanding complex and higher-order graph structures. I am also interested in generative models, graph representation learning, foundation models, higher-order network analysis, and their interdisciplinary applications.

- Topological Deep Learning (TDL)
- Higher-order Network Analysis
- Generative Models
- Graph Representation Learning
- Network Science
- Foundation Models

Education

Imperial College London *London, UK*

PH.D. IN COMPUTER SCIENCE *Sep. 2024 - Present*

- Supervised by Prof. Tolga Birdal
- Awarded the prestigious Lee Family Scholarship (granted to two recipients annually across Imperial)

UESTC (University of Electronic Science and Technology of China) *Chengdu, China*

M.SC. IN COMPUTER SCIENCE *Sep. 2021 - Jun. 2024*

- Supervised by Prof. Linyuan Lü
- GPA: **3.8** / 4.0
- Joint training programs with the University of Science and Technology of China (USTC) and the Yangtze Delta Research Institute (Huzhou)

NUIST (Nanjing University of Information Science and Technology) *Nanjing, China*

B.S. IN COMPUTER SCIENCE *Sep. 2017 - Jun. 2021*

- GPA: **4.1** / 5.0 (**3.9** / 4.0), top **2** in major

Publications

(*: Equal contribution)

Higher-order grammar representations for molecular topology generation and understanding
Yiming Huang, Yujie Zeng, Vijay Prakash Dwivedi, Simone Foti, Jianming Wang, Jure Leskovec, Tolga Birdal
ICLR 2026 Workshop on Foundation Models for Science: Real-World Impact and Science-First Design. [Paper]
TLDR: We introduce HGR, a higher-order grammar representation that converts combinatorial complexes into sequences, enabling direct higher-order topology generation and molecular foundation modeling, together with RingDiv, a ring-enriched benchmark.
Key words: Topological Deep Learning, Generative Model, Molecular Foundation Model

HOG-Diff: Higher-Order Guided Diffusion for Graph Generation
Yiming Huang, Tolga Birdal
International Conference on Learning Representations (ICLR), 2026 [Paper] [Website]
TLDR: We introduce HOG-Diff, a coarse-to-fine graph generation framework that explicitly exploits higher-order topological cues.
Key words: Topological Deep Learning, Generative Model

Cellular-Guided Graph Generative Model
Yiming Huang, Tolga Birdal
ICLR 2025 Workshop on Deep Generative Model in Machine Learning: Theory, Principle and Efficacy, 2025 [Paper]
Key words: Topological Deep Learning, Generative Model

Higher-Order Graph Convolutional Network with Flower-Petals Laplacians on Simplicial Complexes
Yiming Huang*, Yujie Zeng*, Qiang Wu, Linyuan Lü
Proceedings of the AAAI conference on artificial intelligence (AAAI), 2024 [Paper] [Appendix] [Poster]
TLDR: Propose the Flower-Petals (FP) representation and HiGCN, a higher-order GNN with provably stronger expressiveness.
Key words: Topological Deep Learning, Graph Representation Learning, Network Science

Identifying key players in complex networks via network entanglement
Yiming Huang, Hao Wang, Xiao-Long Ren, Linyuan Lü
Communications Physics p. 19. 2024 [Paper]
Key words: Network Science, Influence Maximization

Influential simplices mining via simplicial convolutional networks
Yujie, Zeng*, Yiming Huang*, Qiang Wu, Linyuan Lü
Information Processing & Management p. 103813. 2024 [Paper]
Key words: Topological Deep Learning, Graph Representation Learning, Network Science

Identifying vital nodes through augmented random walks on higher-order networks
Yuejie, Zeng*, Yiming Huang*, Xiao-Long Ren, Linyuan Lü
Information Sciences p. 121067. 2024 [Paper]
Key words: Network Science, Influence Maximization

Cooperative Network Learning for Large-Scale and Decentralized Graphs
Qiang Wu*, Yiming Huang*, Yujie Zeng, Linyuan Lü
IEEE Transactions on Computational Social Systems. 2026 [Paper]
Key words: Graph Representation Learning, Network Science

A novel coherence-based quantum steganalysis protocol
Zhiguo Qu, Yiming Huang, Min Zheng
Quantum Information Processing p. 362. 2020 [Paper]
Key words: Quantum Computing

An efficient logic-guided attack strategy on logical-physical coupled networks
Mengying Geng, Ling Zhang, Min Hui, Yiming Huang, Feng Qing, Mengqiao Xu, Changjun Fan, Guangquan Cheng
Chaos, Solitons and Fractals 202. p. 117396. 2026 [Paper]

Projects

Topological Deep Learning and Generative Models — *Imperial, UK*
Sep. 2024 - Present

- Designing diffusion-based graph generative models guided by higher-order topological structures.
- Developing molecular representations that explicitly encode higher-order structural patterns and generalize well to downstream tasks.
- Constructing benchmarks with intricate molecular topologies and advancing scalable molecular/graph foundation models.

Graph Representation Learning and Higher-order Network Analysis — *Yangtze Delta Region Institute (Huzhou) & UESTC, China*
Sep. 2022 - Jun. 2024

- Developed higher-order Laplacians and proposed higher-order graph neural networks with superior expressiveness.
- Contributed to the Chinese monograph *Graph Machine Learning*; collaborated with Prof. Jure Leskovec.

Network Robustness and Influence Maximization Analysis — *UESTC, China*
Sep. 2021 - Sep. 2022

- Investigated network robustness and influence dynamics using higher-order structures and physics-inspired models.

Honors & Awards

Total: 1 international award, more than **10** provincial and above awards; **7** patents; **1** national research project.

Awards

2023	**Gold Prize**, "Internet+" College Students Innovation and Entrepreneurship Competition (**Top 1%**)	
2021	**Honor Graduate**, (Top 10%)	*NUIST*
2020	**Finalist**, Mathematical Contest In Modeling (MCM) (**Top 1%**)	
2020	**Project Leader**, National Training Program of Innovation and Entrepreneurship for Undergraduates	

Scholarships

2024-2028	**Lee Family Scholarship**, (**Top 2**, two recipients annually across Imperial)	*Imeprial*
2021-2023	**First-Class**, Academic Scholarship of Master Degree student (Top 10%, received each academic year)	*UESTC*
2017-2020	**First-Class**, Academic Scholarship of Bachelor Degree student (Top 10%, received each academic year)	*NUIST*

Academic Services

Conference Reviewing	NeurIPS (2025), ICML (2025, 2026), AAAI (2025), ICLR (2025, 2026), ECCV (2026)
Journal Reviewing	Nature Communications, Neurocomputing, Entropy, Applied Sciences, Journal of Computational Science, IEEE Transactions on Artificial Intelligence (TAI), IEEE Transactions on Computational Social Systems (TCSS), Physica A: Statistical Mechanics and its Applications, International Journal of Modern Physics C
Organizing Committee	LOGML (London Geometry and Machine Learning) Summer School 2026 [Website]